FraudScope

Fraud Detection with Temporal Graph Neural Networks

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Applied machine learning portfolio
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Synthetic research and a deployed private workbench

Fraud review under limited capacity

5% review budget

Rank transactions so analysts can focus their attention.

Measure fraud recall and value capture at the same review budget.

The operating tradeoff

False positives consume analyst time and may inconvenience customers.

A high ranking score does not automatically imply the lowest operating cost.

The 5% budget and cost assumptions define the experiment, not an employer's operating policy.

End-to-end application

Layer	Working behavior
Analyst interface	Review queue, case details, graph evidence, model comparison
API	Input validation, ownership, idempotency, revision checks
Inference	Calibrated logistic decisions and a GNN in shadow mode
Persistent state	Cloudflare D1 records, reviews and audit chain
Research dashboard	Seven model families and frozen structural stress results

React and TypeScript on Cloudflare Workers. The current deployment is owner-private.

Chronological evaluation protocol

Stage	Elapsed-time allocation	Purpose
Training	55%	Fit preprocessing and model parameters
Calibration	15%	Map scores to probability estimates
Validation	10%	Select model and decision threshold
Final test	20%	Evaluate the frozen choices

Three seeds, 60,000 events per seed. A two-day assumed label-maturity purge applies at boundaries.

The temporal graph contract

Element	Implementation
Prediction unit	A transaction at time t
Account relation	Earlier transactions on the same account, within 7 days
Device relation	Earlier transactions on the same device, within 1 day
Neighborhood bound	Latest 8 neighbors per relation
Leakage rule	Exclude current, equal-time and future events

One-hop, relation-typed transaction projection. This implementation has no recurrent TGN memory.

Modern GNN ideas with explicit controls

Graph candidates

Relational GraphSAGE-style aggregation

Temporal multi-head relation attention

Risk-gated temporal attention

Experimental controls

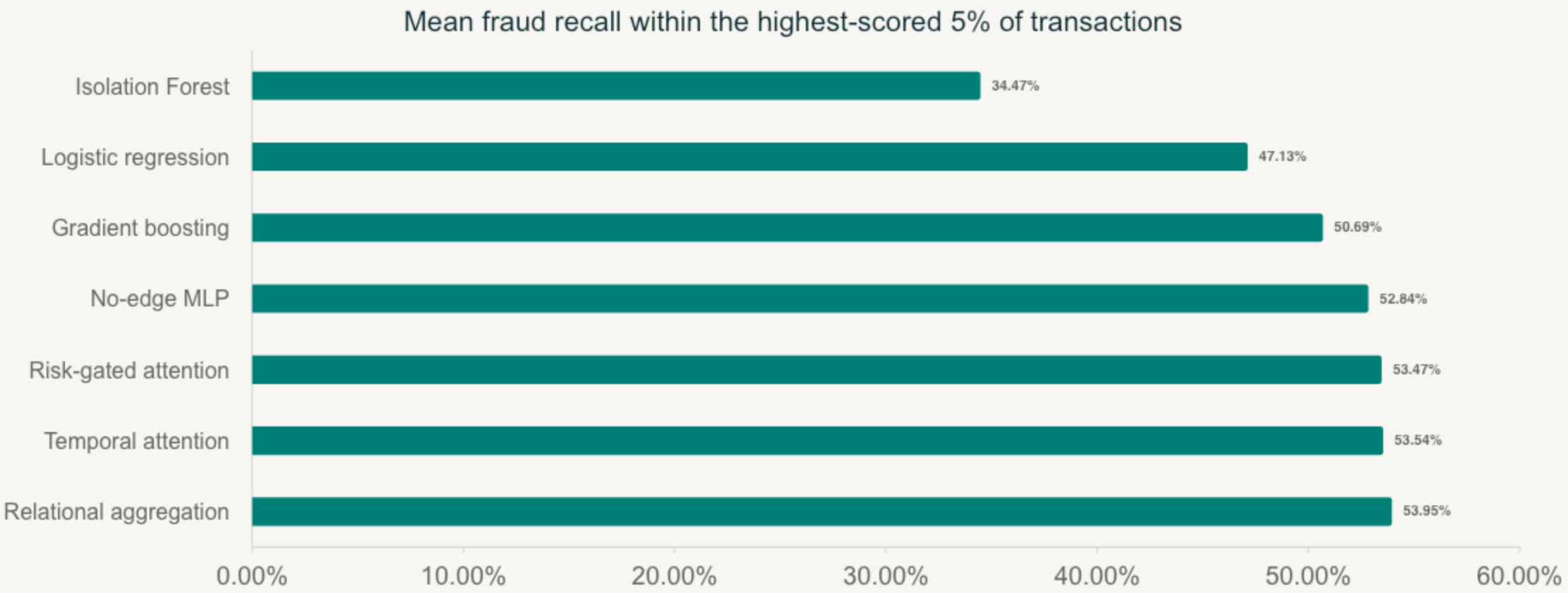
No-edge MLP with the same encoded features

Logistic regression, gradient boosting and Isolation Forest

Matched data and chronological splits

Risk gating adapts a 2026 preprint using a causal logistic teacher. It is not an exact paper reproduction.

GraphSAGE leads recall at the 5% budget



Synthetic means across three seeds. GraphSAGE exceeds the no-edge MLP by 1.11 percentage points.

Average precision differences are modest

Model	Mean AP	Sample SD
Logistic regression	0.4516	0.0206
Gradient boosting	0.4855	0.0233
Isolation Forest	0.3422	0.0148
No-edge MLP	0.4942	0.0266
Relational GraphSAGE	0.4974	0.0235
Temporal attention	0.4929	0.0268
Risk-gated attention	0.4941	0.0251

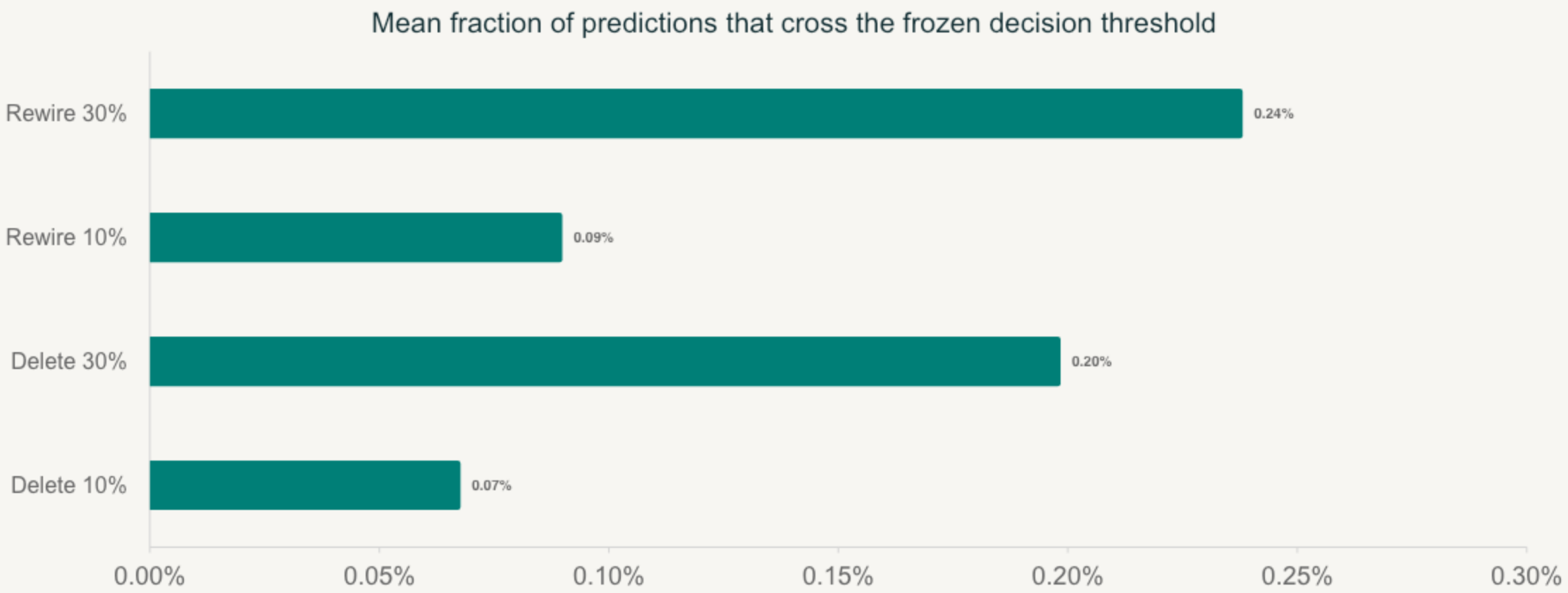
Three synthetic seeds. Sample standard deviation describes seed variation and is not a confidence interval.

Simpler models win the validation cost rule

Seed	Overall selected model	Graph-only selection
42	No-edge MLP	Temporal attention
73	No-edge MLP	Temporal attention
101	Gradient boosting	Temporal attention

The live app retains the original calibrated logistic policy. GNN comparison remains in shadow mode.

Structural stress measures decision sensitivity



Historical three-seed chart means. Raw multi-seed stress files remain pending recovery. Weights and policy were frozen.

Serving reliability has executable evidence

State invariants

Exact retries return the stored response.

Conflicting payloads and stale reviews fail.

Workspace ownership scopes every case.

Model and audit checks

80 independent model fixtures
48 raw-event replay cases

Maximum calibrated GNN error: 0.00000158

Internal audit verification detects tested record changes.

Fresh checks: 12 application tests. Mac Python research: 41 passed, one PyTorch module skipped.

Three-minute employer demonstration

Action	Evidence to explain
Replay the synthetic feed	24 persisted events and saved model decisions
Open a case and its graph	Only historical account/device neighbors
Compare models and save a review	Shadow scores, analyst notes and revision control
Verify audit and export records	Consistent case history and inspectable CSV
Open the research dashboard	Ranking gains, cost selection and stress limits

Owner screen share for the private app. The local reviewer tour is the credential-free fallback.

The path to real-data deployment

Gate	Required evidence
Data authorization	Permitted IEEE-CIS inputs and documented data lineage
Feature validity	Entity proxy quality, availability at scoring time, label delay
Model evaluation	Frozen temporal protocol, calibration, subgroup and cost analysis
Operational readiness	Target-scale load, recovery drill, monitoring and accountable owners
Release approval	Independent security and model-risk review, controlled rollout

Real IEEE-CIS benchmarking has not run. No payments are connected or automatically blocked.

Research judgment and engineering ownership

What I can demonstrate

Explain causal graph construction and leakage controls.

Reproduce model comparisons and defend selection tradeoffs.

Trace an event through scoring, review and audit.

What I would extend next

Evaluate authorized real data.

Measure graph gains against stronger tuned baselines.

Study stability under realistic entity-link errors and drift.

AI-assisted reference project. Present these capabilities after independently reproducing and mastering the code.

Appendix: verification commands

From applications/workbench	Expected evidence
pnpm install --frozen-lockfile	Declared dependency set
cp .openai/hosting.example.json .openai/hosting.json	Local configuration without a Site ID
pnpm test	12 passing application tests
pnpm run check	TypeScript exits successfully
pnpm run review:verify	24 events and 25 valid audit records

Node 22.13+ and pnpm 11.25.0. Follow docs/HIRING_MANAGER_REVIEW.md for clone, pull and research reproduction.

Appendix: sources and evidence

Project metrics: reports/gnn/multi_seed_summary.json

Stress chart recovery: reports/recovery/structural_chart.json

Serving parity: applications/workbench/model-data/parity-report.json

RGH-GNN, Hari et al., 2026 preprint: arxiv.org/abs/2610.27107

BRIGHT, Lu et al., CIKM 2022: arxiv.org/abs/2205.13084

Temporal Graph Networks, Rossi et al., 2020: arxiv.org/abs/2006.10637

Project evidence supports synthetic findings only. Research sources inspire the design and do not validate this implementation.